Logic Minimization – Part 2

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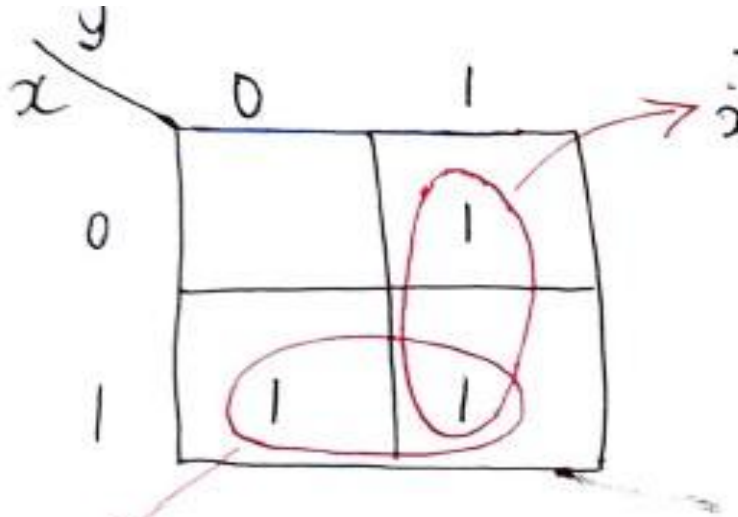
Review

- We have discussed about minterm expansion and incompletely specified functions.
- In this lecture, we will discuss a graphical method of simplification.

Motivation

- The algebraic method discussed so far is somewhat adhoc since there are no specific rules to predict the successive steps.
- We will discuss a graphical method based on what is called Karnaugh Map (or K-map) to handle simplification of Boolean functions of two or more variables.
- The goal is to get a simplified sum-of-products (SOP) form

K-map for xy' + x'y + xy (2-variable case)



x gets eliminated
Remainig is y

y gets eliminated
Remainig is x

Hence Simplified function is x + y

Explanation

- Put a 1 in the square corresponding to a term in the given function.
- Then we combine pairs of adjacent 1 cells depending on what is given as input. This allows us to eliminate variables. For example, the top ellipse/loop (in the previous slide) corresponds to x'y + xy and we note x changes from 0 to 1 as we move down – this means variable x will be eliminated. Similarly, for the other loop (horizontal).

How do we know the solution is correct ?

- We have shown via the K-map that xy’ + x’y + xy simplifies to x + y.
- How can you verify this ?
- Question for students

Verification

$$f = xy' + x'y + xy$$

$$= xy' + x'y + \underline{xy} + \underline{xy}$$

(can always add a term that is in the list)

$$= xy' + xy + x'y + xy$$

(moving one xy inside)

$$= x(y+y') + y(x'+x)$$

$$= x + y$$

Remark

- In general, loops in a K-map will cover cells which happen to be a power of 2 (i.e, 2^1 , 2^2 , 2^3 etc.).
- That is, adjacent 1's are covered but only if we have two adjacent 1's, four adjacent 1's, eight adjacent 1's etc. A sole 1 somewhere (without any adjacent 1) is also important (and needs to be part of the final expression).
- Think of the reason for not having a loop around three adjacent 1's. We'll next discuss 3-variable K-maps but prior to that a note on conventions followed.

Convention on location of minterms

Mano - Ciletti

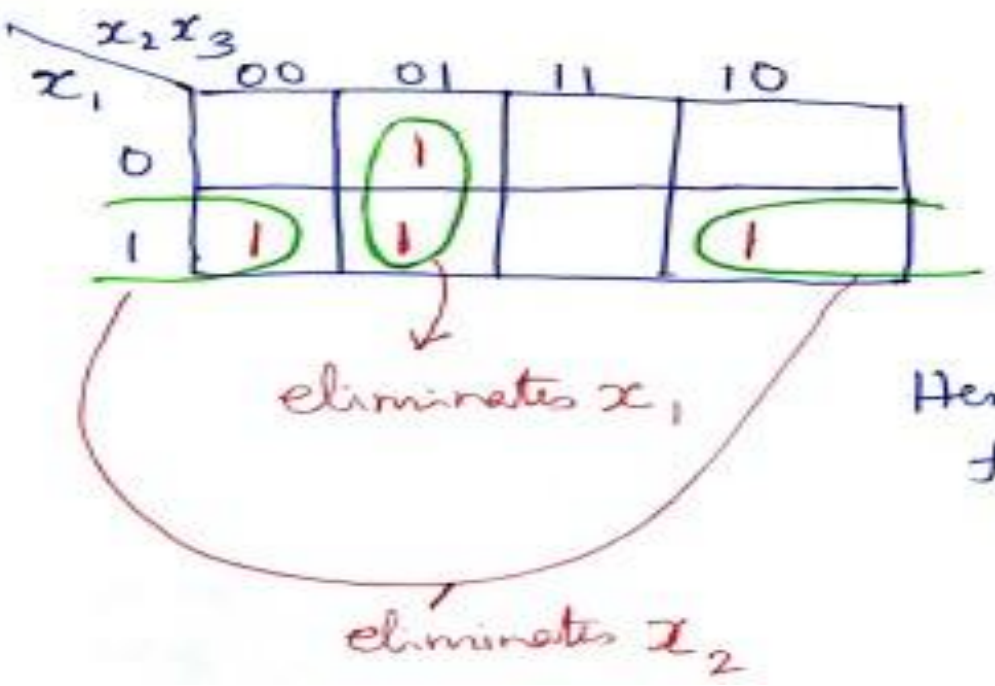
a \ bc	00	01	11	10
0	0	1	3	2
1	4	5	7	6

Roth, Wakerly etc.

bc \ a	0	1
00	0	4
01	1	5
11	3	7
10	2	6

K-map for a three-variable function

$$\bar{x}_1 \bar{x}_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3$$



Hence, simplified form is $x_1 \bar{x}_3 + \bar{x}_2 x_3$

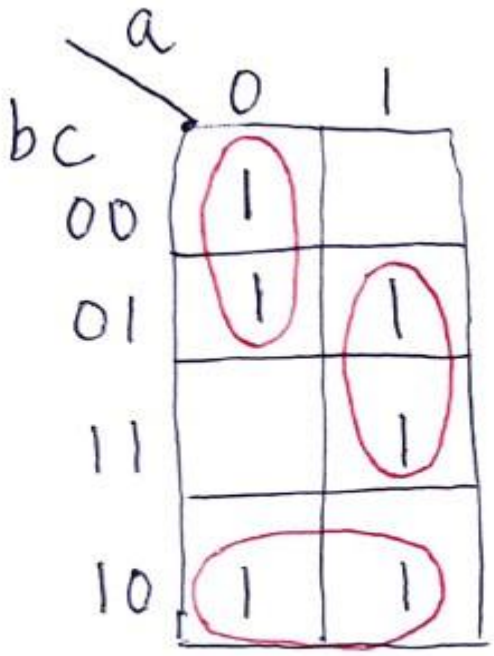
Explanation

- Note the pattern in the code at the top. We use Gray code sequence to allow us to perform simplification by combining adjacent 1’s.
- Also, note that the left bottom cell (corresponding to 001) and the right bottom cell (corresponding to 101) are adjacent.

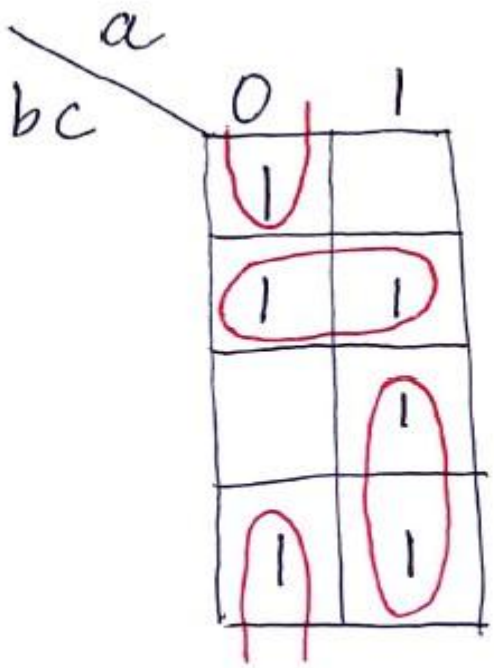
Function with two min SOP forms

Consider

$F(a,b,c) = \sum (0,1,2,5,6,7)$



$F = a'b' + bc' + ac$



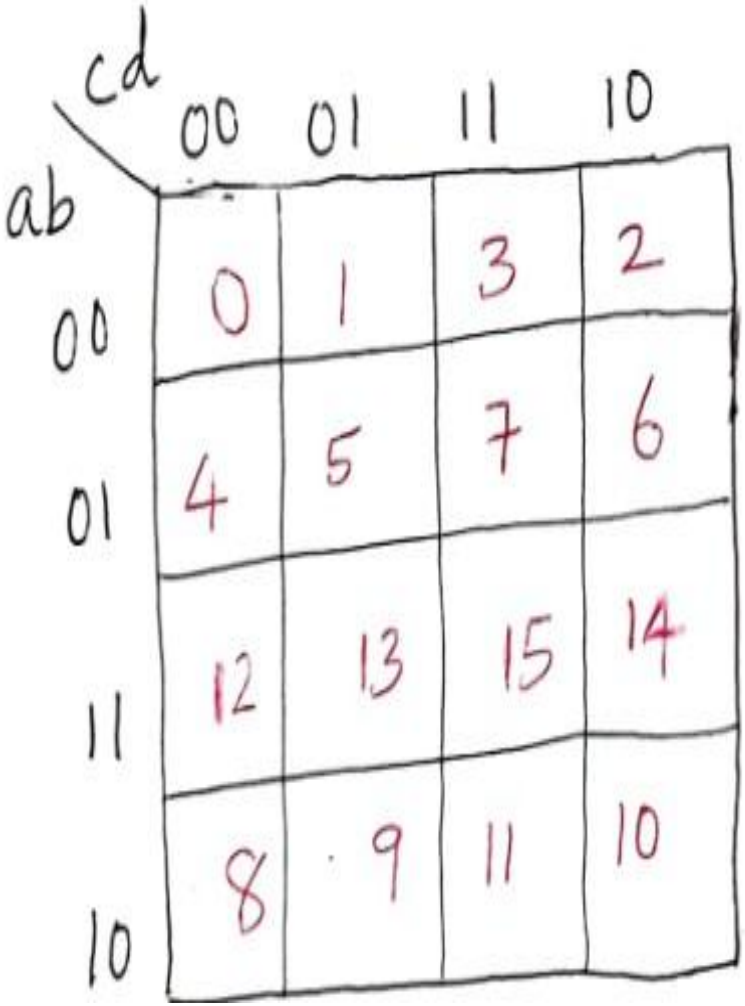
$F = a'c' + b'c + ab$

Beyond 3 variables

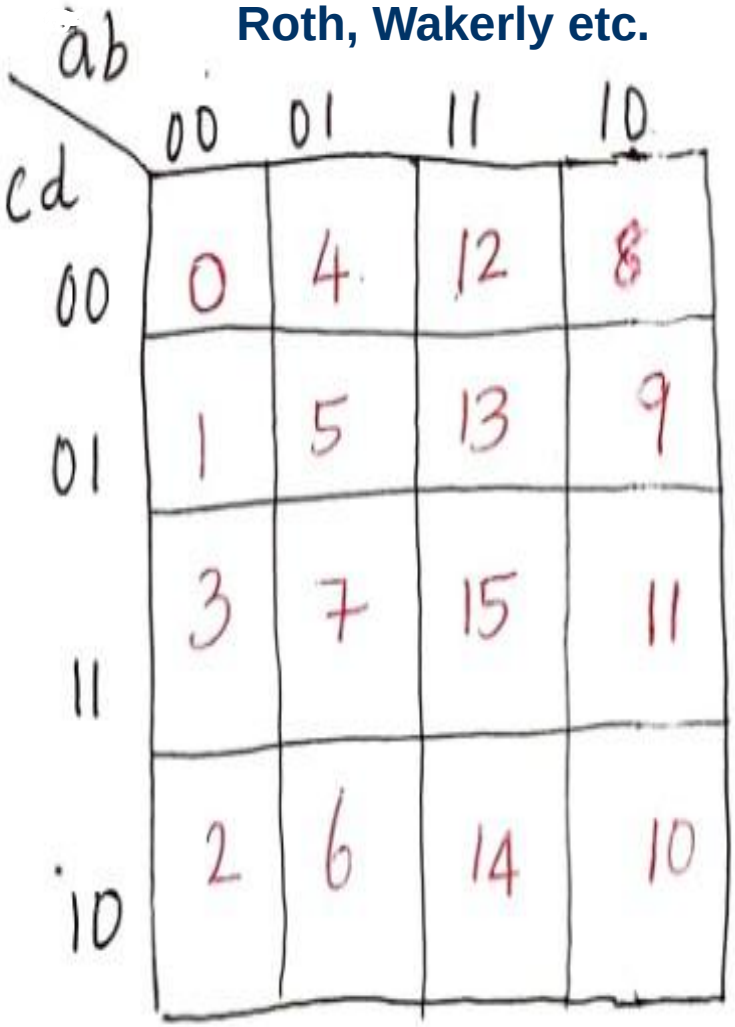
- The K-map can be used for simplifying functions of 4 variables too.
- Before giving an example, we will indicate the convention followed by different books regarding the location of minterms.
- We depict the convention pictorially next.

Convention regarding location of minterms

Mano-Ciletti



Roth, Wakerly etc.



Example for four variables

- Consider the function

  $f = \sum (1,3,5,10,12,13)$.

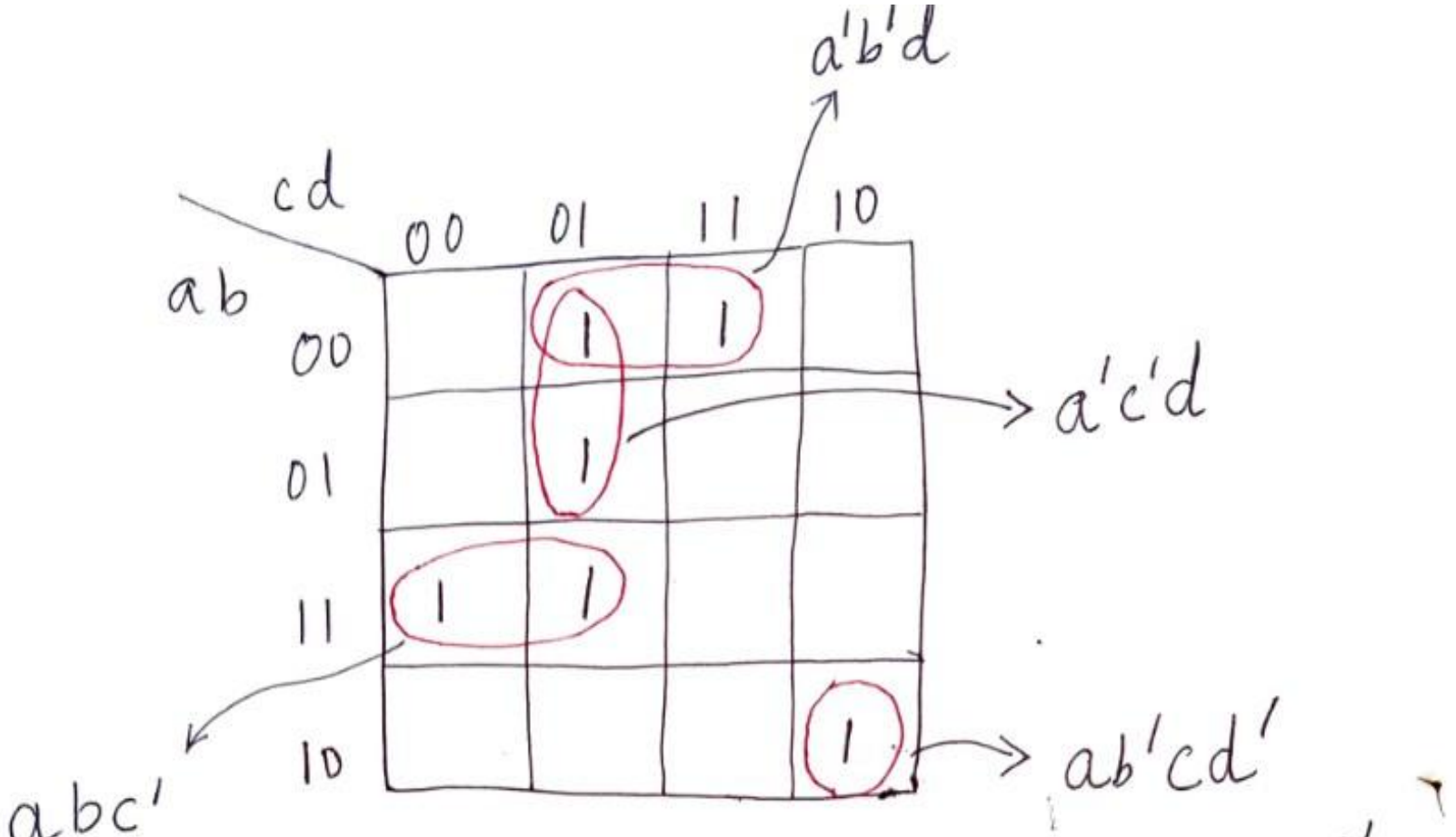
  (this may be sometimes written also as

  $f = \sum m(1,3,5,10,12,13)$)

The K-map will now have 16 cells. Gray code is used as before to label at the top (and on the side).

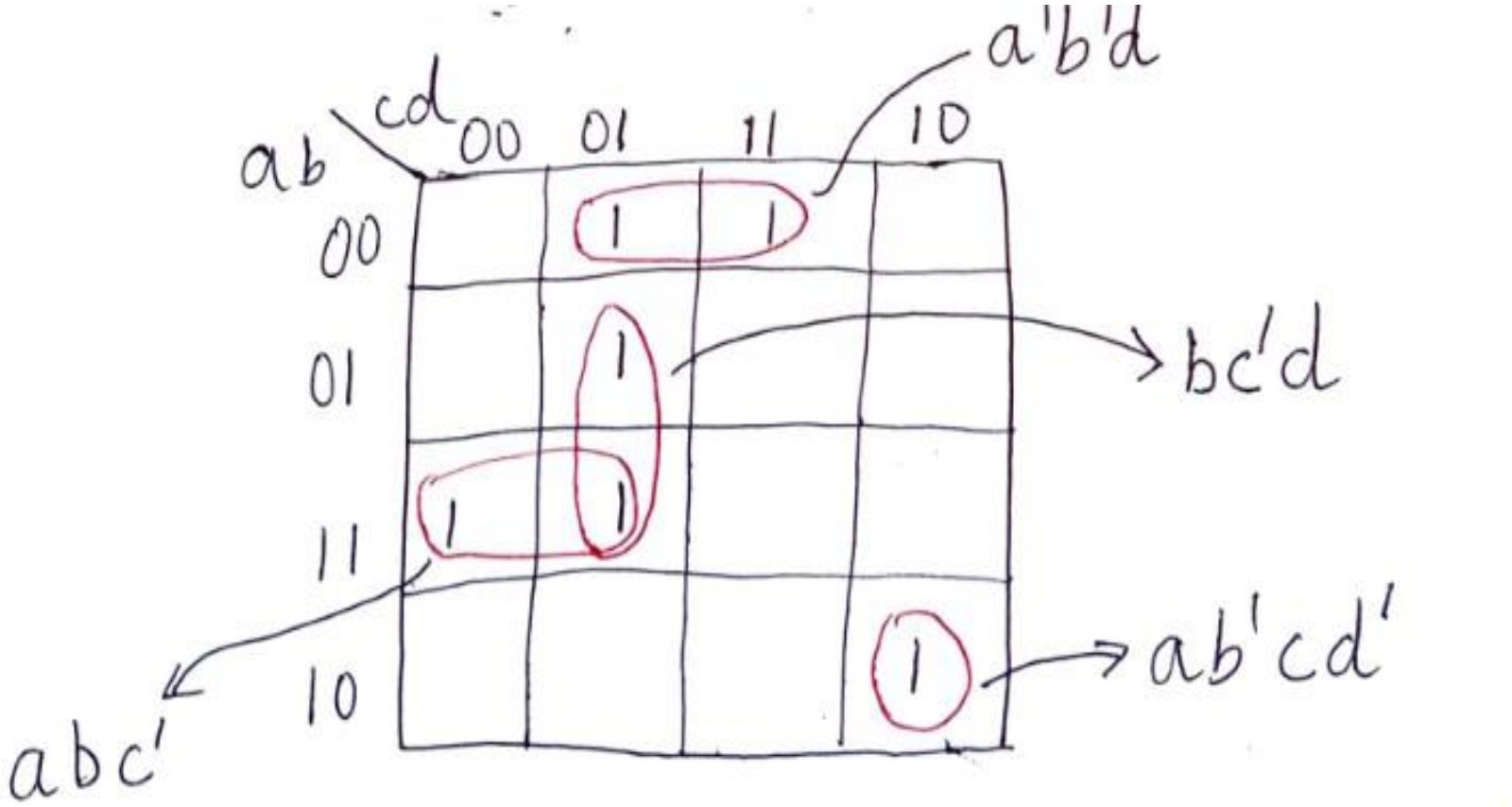
We illustrate the solutions next.

Solution 1



$$f = abc' + a'b'd + a'c'd + ab'cd'$$

Solution 2



$$f = abc' + a'b'd + bc'd + ab'cd'$$

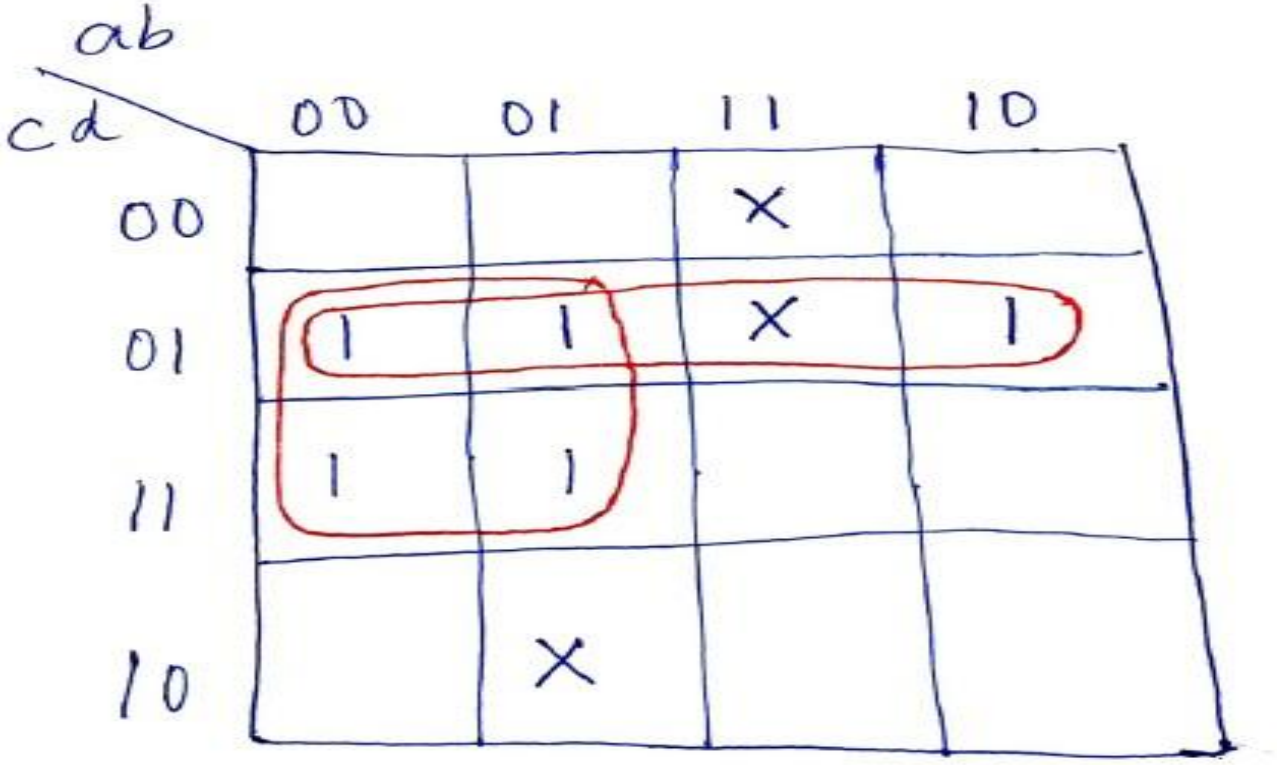
Simplification of an incompletely specified function

- Consider a function F given by

$$F = \sum m(1,3,5,7,9) + \sum d\,(6,12,13).$$

We want to simplify this. Note that the minterms are given by m(1,3,5,7,9) while the don't care terms are given by d (6,12,13). The K-map is shown next.

The K-map for the incompletely specified function



Hence, the solution is a'd + c'd

Explanation

- Note that all the minterms need to be used.
- The don’t care terms are used only if they will help in simplification. Hence, the K-map does not have loops covering all the don’t care terms.
- Note that the only don’t care term that helps in getting a simplified expression is 13.

Can we go beyond 4 variables ?

- Yes, one can have K-maps for 5 variables. We need to have two 4-variables K-maps, look at overlaps and so on. We will skip this.
- Beyond 5 variables, one needs to look at other methods of simplification (for example, there is something called as Quine-McCluskey (QM) method based on prime implicant charts).
- We will not explicitly discuss QM method but will look at prime implicants using K-maps.

What is a practical application of a K-map ?

- K-maps can be used in design of control units for certain microprocessors.
- In particular, they are useful in design of hardwired control units for contemporary instruction set processors (eg. ARM-based processors used in mobiles).

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- C. Roth and L.L. Kinney, Fundamentals of Logic Design, Gamma Publications, 2021